

Manoj Kumar Eede

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Professional Summary

Software Engineer with 2+ YOE in scalable backend systems, distributed microservices, and AI-powered platforms. Delivered 0-to-1 initiatives powering PDF intelligence in ChatGPT and handwritten processing for Adobe AI applications, serving 300K+ enterprise transactions. Expertise in software development, ML, and GenAI (RAG, LLMs, NLP) in Java, Python, and cloud-native technologies.

Technical Skills

Languages: Java, Python, C++, JavaScript, Swift, SQL

Libraries & Frameworks: Spring Boot, Flask, React, LangChain, PyTorch, TensorFlow, Scikit-learn

Databases & Architecture: MySQL, PostgreSQL, Redis, Microservices, REST APIs, Distributed Systems, RAG Pipelines

Cloud & DevOps: Azure, Kubernetes, Docker, Argo CD, GitHub Actions, Splunk, Prometheus, CI/CD, MLOps

Work Experience

Adobe

Noida, India

Member of Technical Staff II

Jan 2026 – Present

- Building scanned page classifier for handwritten vs. printed document detection using CNNs, Vision Transformers, and transfer learning; enabling intelligent routing across downstream AI services.
- Architected an AI/ML-powered PDF digitization service leveraging computer vision and MLLMs to convert handwritten documents into structured digital formats, powering handwriting capabilities in Adobe AI applications (Study Spaces, Adobe Scan).
- Orchestrated external rollout of PDF conversion backend services as a Java REST API; engineering multi-language SDKs (Java, Python, .NET, Node.js), projected to generate **\$1M ARR** in the initial phase.

Member of Technical Staff

Sept 2024 – Jan 2026

- Architected 0-to-1 production-grade PDF-to-Markdown and Markdown-to-PDF pipelines across distributed systems (Python OCR engine, Java microservices, React playground with token-based throttling); integrated into **ChatGPT's** Acrobat SDK (MCP tools) for AI-assisted document workflows at scale.
- Owned feature development for Adobe's core Java-based DocGen API, reducing error rates to **<0.01%** across **300K+** monthly enterprise transactions.
- Built a GenAI-powered financial document assistant using an end-to-end RAG pipeline for extraction, reasoning, and insights.
- Tech Stack:** Java, Python, React, Argo CD, Kubernetes, Docker, Splunk, Prometheus, Redis, LangChain

Asper.ai

Bangalore, India

Data Scientist

June 2024 – Sept 2024

- Led end-to-end client engagement for a **\$3M** contract with a leading US-based CPG company (1,000+ SKUs), spanning EDA, demand forecasting algorithm design, and post-production RCA; improved demand planning accuracy by over **10%**.
- Designed and implemented scalable ML system design frameworks to improve efficiency and model deployment.
- Tech Stack:** Python, TensorFlow, Scikit-learn, Azure, Kubernetes, MLOps

Adobe Research Lab

Bangalore, India

Research Intern

May 2023 – July 2023

- Fine-tuned Flan-T5 and used BERT for Controlled and Grounded Text Generation to enhance reviewer efficiency.
- Developed an Automated Commenting Pipeline using advanced NLP, hybrid embeddings (semantic + lexical), and prompt engineering, achieving **>15%** higher accuracy on curated NLPeer data; integrated into Adobe's GenAI framework.
- Tech Stack:** Python, PyTorch, Flask, React

Education

Indian Institute of Technology, Roorkee

Roorkee, India

Integrated Masters in Mathematics and Computing (Applied Mathematics)

July 2019 – May 2024

CGPA: 8.78 / 10 Coursework: OOP, DBMS, DSA, OS, Discrete Mathematics, Graph Theory, Digital Image Processing

Projects

FinGen – Agentic-AI Powered Investment Assistant

2025 – 2026

Built a full-stack agentic AI platform for Indian stock market insights, sentiment analysis, and portfolio tracking. Deployed on Hugging Face Spaces with load balancing, caching, and CI/CD. *Stack: PyTorch, SmolAgents, Transformers, Supabase, Streamlit.*

CommerceHub – Cloud-Native E-Commerce Platform (GitHub)

2024

Built a cloud-native e-commerce platform with JWT auth, React frontend, and Spring Boot backend with Redis caching, PostgreSQL, and unit testing. *Stack: Java, Spring Boot, React, Redis, PostgreSQL, Docker, JUnit.*

Publications

Demand Forecasting Using Statistical Machine Learning Models.

[Springer Nature, forthcoming '26]

Madhu Jain, Eede Manoj Kumar, Nidhi Sharma.

In *Advanced Machine Learning and Evolutionary Computing Algorithms for Financial Decision-Making*.

Awards and Achievements

- SPOT Award (2025):** Recognized by Adobe for outstanding contributions to AI/ML initiatives.
- INSPIRE Scholarship, DST MHRD (2019–2024):** Awarded to top 1% students nationwide in pure sciences.
- Competitive Programming:** Codeforces – **1663** | **Expert** (Handle: codehacker4689), LeetCode – **2047** (Handle: 19312009).
- Open Source – Docx4j:** Identified and resolved a critical DOCX corruption bug; fix merged into the library's release. (GitHub)

Certifications

AI Agents (Hugging Face) · Model Context Protocol (Hugging Face) · Time Series Analysis, Forecasting & ML (Udemy) · Deep Learning Specialization (Udemy) · ML Regression Master Class in Python (Udemy)